MIL-M-38510/61B
AMENDMENT 2
24 September 1998
SUPERSEDING
AMENDMENT 1
15 May 1997

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, ECL, FLIP-FLOPS, MONOLITHIC SILICON

INACTIVE FOR NEW DESIGN AFTER 6 September 1996

This amendment forms a part of MIL-M-38510/61B,
dated 15 July 1985, and is approved for used by
all Departments and Agencies of the Department of Defense.

The attached insertable replacement pages listed below are replacements for stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

Replacement page	Page replaced
13	reprinted without change
14	new page
15	reprinted without change
16	new page
17	new page
18	new page

PAGE 21

FIGURE 10, after "Q OUTPUT" waveform and before "GENERATOR CHARACTERISTICS", insert:

"FREQ. (OUT) = $\frac{\text{FREQ. (IN)}}{2}$

F_{MAX} IS HIGHEST INPUT FREQUENCY AT WHICH DEVICE CEASES TO TOGGLE"

PAGE 32

* TABLE III, F_{MAX} row, Min column of subgroup 9, delete "125" and substitute "62.5" in two places.
(This reverses a change made by amendment 1.)

* TABLE III, F_{MAX} row, Min column of subgroup 10, delete "125" and substitute "62.5" in two places.
(This reverses a change made by amendment 1.)

* TABLE III, F_{MAX} row, Min column of subgroup 11, delete "105" and substitute "52.5" in two places.
(This reverses a change made by amendment 1.)

PAGE 34

TABLE III, I_{IH3}, Max column of subgroup 2, delete "420" and substitute "375" in two places.

TABLE III, I_{IH4}, Max column of subgroup 2, delete "450" and substitute "495" .

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PAGE 35

* TABLE III, F_{MAX} row, Min column of subgroup 9, delete “200” and substitute “100” in two places. (This reverses a change made by amendment 1.)

* TABLE III, F_{MAX} row, Min column of subgroup 10, delete “200” and substitute “100” in two places. (This reverses a change made by amendment 1.)

* TABLE III, F_{MAX} row, Min column of subgroup 11, delete “200” and substitute “100” in two places. (This reverses a change made by amendment 1.)

PAGE 37

* TABLE III, F_{MAX} row, Min column of subgroup 1, delete “125” and substitute “62.5” in two places. (This reverses a change made by amendment 1.)

* TABLE III, F_{MAX} row, Min column of subgroup 2, delete “125” and substitute “62.5” in two places. (This reverses a change made by amendment 1.)

* TABLE III, F_{MAX} row, Min column of subgroup 3, delete “115” and substitute “57.5” in two places. (This reverses a change made by amendment 1.)

PAGE 43

* TABLE III, F_{MAX} row, Min column of subgroup 9, delete “125” and substitute “62.5” in two places. (This reverses a change made by amendment 1.)

* TABLE III, F_{MAX} row, Min column of subgroup 10, delete “115” and substitute “57.5” in two places. (This reverses a change made by amendment 1.)

* TABLE III, F_{MAX} row, Min column of subgroup 11, delete “125” and substitute “52.5” in two places. (This reverses a change made by amendment 1.)

The margins of this amendment are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous amendment.

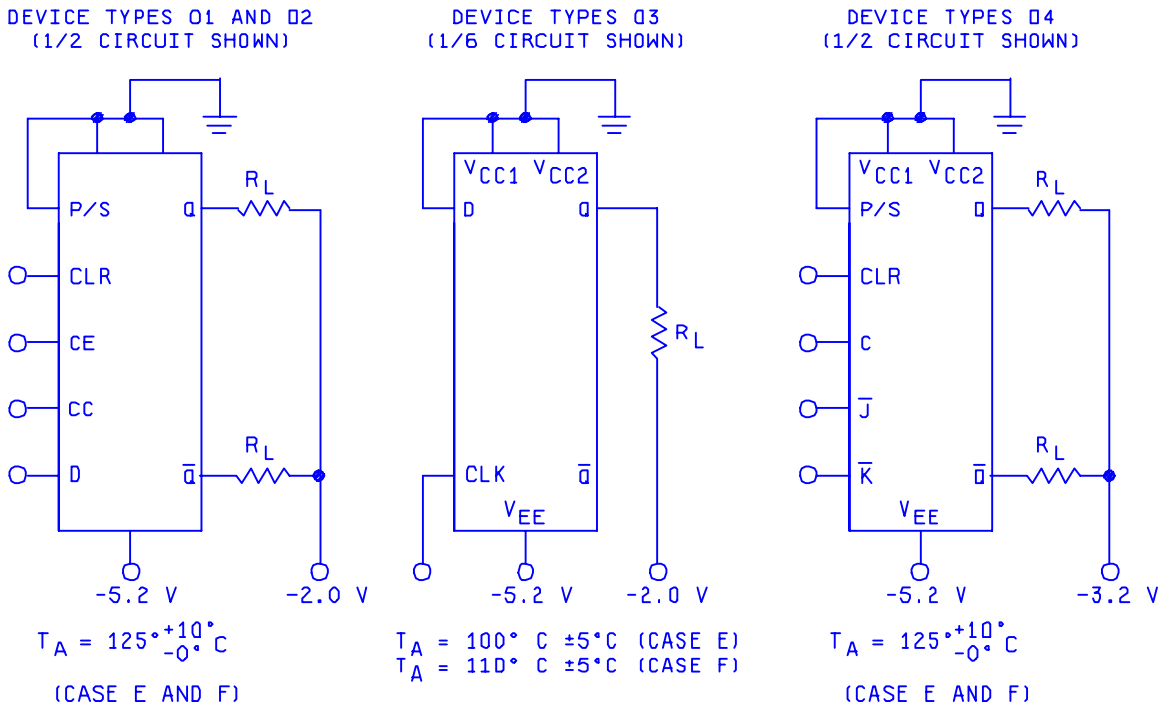
Custodians:
Army - CR
Navy - EC
Air Force - 17

Preparing activity:
DLA - CC

(Project 5962-1828)

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC, OS, SH
Air Force - 19, 85, 99

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NOTES:
1. R_L = 100Ω ± 5% rated at 70 MW at 125° C (or equivalent).
2. Momentary ground on clock input (device type 03) after V_{EE} and -2.0 V have been applied.
3. Inputs are internally returned to V_{EE} through 50 kΩ resistors.

FIGURE 3. Burn-in and life test circuits.

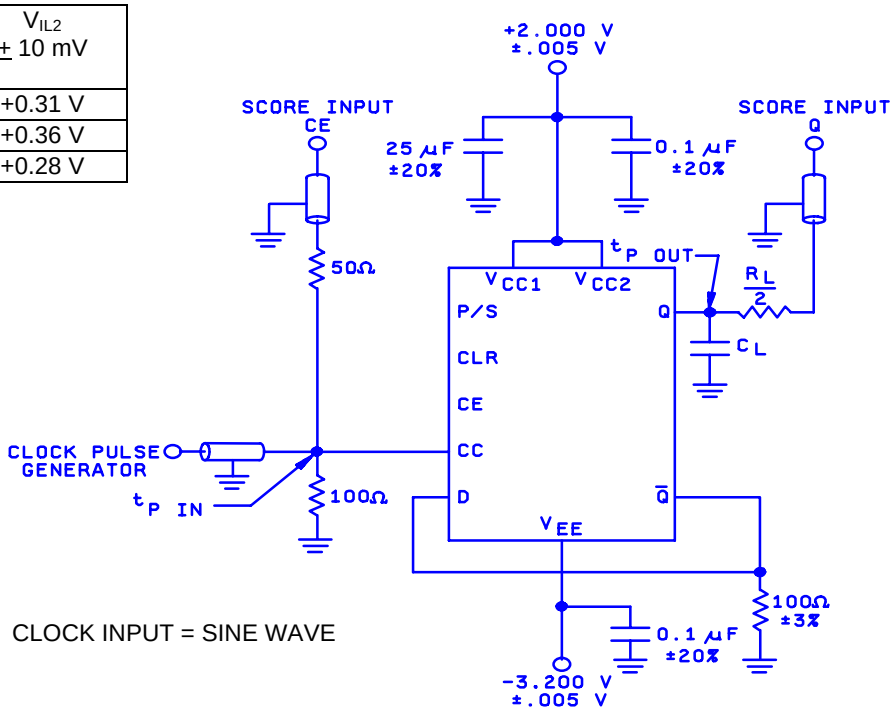
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GENERATOR CHARACTERISTICS

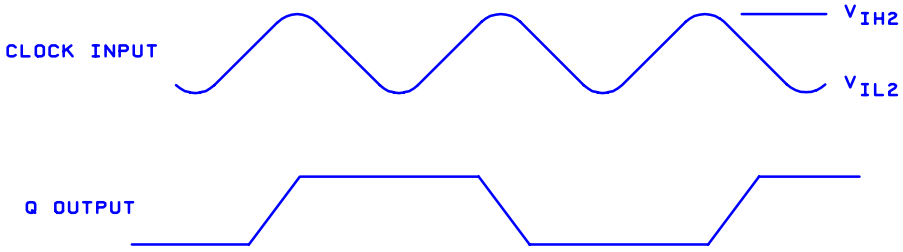
T_C	V_{IH2} ± 10 mV	V_{IL2} ± 10 mV
25 °C	+1.11 V	+0.31 V
125 °C	+1.24 V	+0.36 V
-55 °C	+1.01 V	+0.28 V

$\frac{R_L}{2} = 50\Omega \pm 5\%$

SCOPE INPUT = 50Ω to GND
C_L (test jig) ≤ 5 pF



CLOCK INPUT = SINE WAVE



$$\text{FREQUENCY (OUT)} = \frac{\text{FREQUENCY (IN)}}{2}$$

FMAX IS HIGHEST INPUT FREQUENCY AT WHICH DEVICE CEASES TO TOGGLE

NOTES:
1. Perform test in accordance with test table; each output is tested separately.
2. All input and output cables are equal lengths of 50 ohm coaxial cables. Wire length should be ≤ .250 (6.35 mm) from t_p in to input pin and t_p out to output pin.
3. Outputs not under test connected to a 100 ohm resistor to ground.
4. Note that observed pulse amplitude is attenuated by one half.

FIGURE 4. F_{MAX} test circuit for device types 01 and 02.

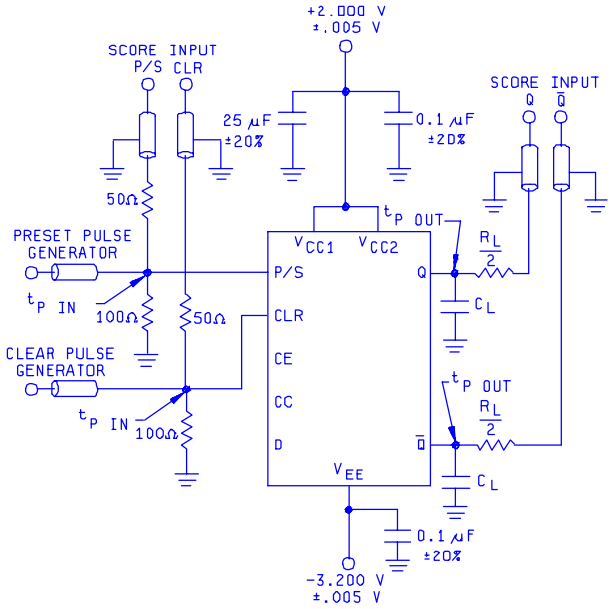
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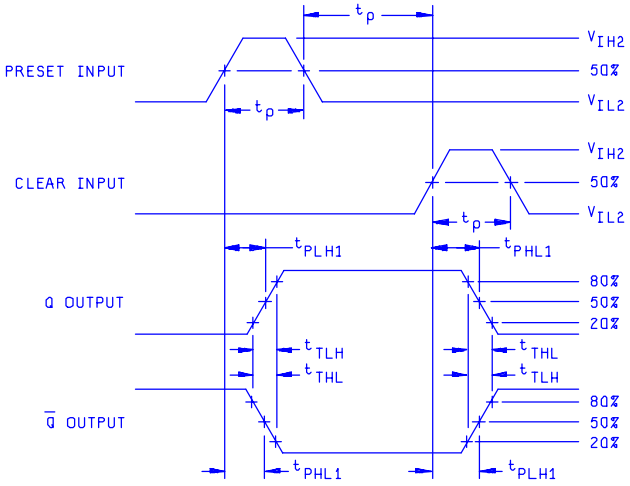
GENERATOR CHARACTERISTICS

T_C	V_{IH2} ±10 mV	V_{IL2} ±10 mV
25°C	+1.11 V	+0.31 V
125°C	+1.24 V	+0.36 V
-55°C	+1.01 V	+0.28 V

Z_{OUT} = 50Ω
t_P = (P/S & CLR) = 40 ns
PRR = 1 MHz
Device type 01
t_{THL} = 2.0 ns (20% - 80%)
t_{TLH} = 2.0 ns (20% - 80%)
Device type 02
t_{THL} = 2.0 ns (20% - 80%)
t_{TLH} = 2.0 ns (20% - 80%)

$R_L/2$ = 50Ω ± 5%
SCOPE INPUT = 50Ω TO GND
C_L (test jig) ≤ 5 pF





NOTES:
1. Perform test in accordance with test table; each output is tested separately.
2. All input and output cables are equal lengths of 50Ω coax cables. Wire length should be ≤ .250 (6.5 mm) from t_P in to input pin and t_P out to output pin.
3. Outputs not under test connected to a 100 ohm resistor to ground.
4. Note that observed pulse amplitude is attenuated by one half.

FIGURE 5. Preset and clear switching test circuit for device types 01 and 02.

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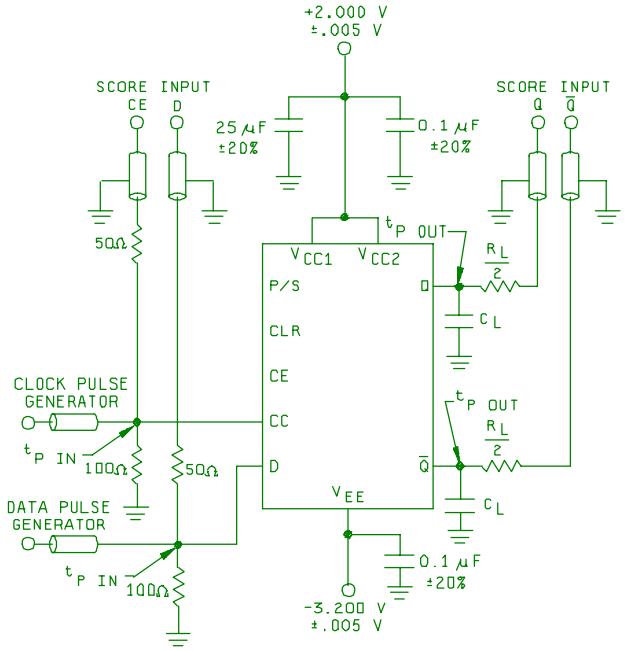
GENERATOR CHARACTERISTICS

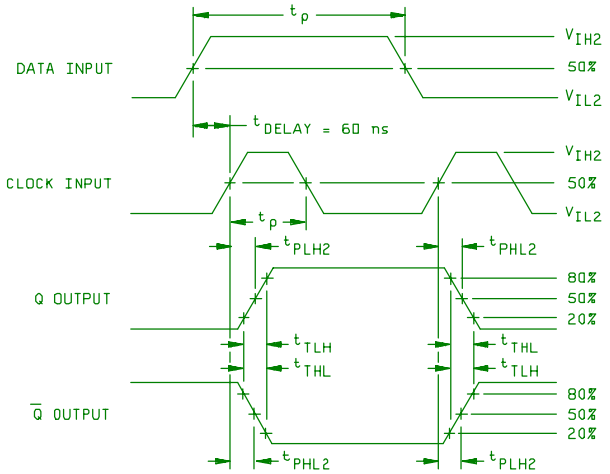
T_C	V_{IH2} ± 10 mV	V_{IL2} ± 10 mV
25 °C	+1.11 V	+0.31 V
125 °C	+1.24 V	+0.36 V
-55 °C	+1.01 V	+0.28 V

Z_{OUT} = 50Ω
t_p (data) = 150 ns
t_p (clock) = 40 ns
PRR = 1 MHz
Device type 01
T_{THL} = 2.0 ns (20%-80%)
T_{TLH} = 2.0 ns (20%-80%)
Device type 02
T_{THL} = 1.5 ns (20%-80%)
T_{TLH} = 1.5 ns (20%-80%)

$\frac{R_L}{2}$ = 50Ω ± 5%

SCOPE INPUT = 50Ω to GND
C_L (test jig) ≤ 5 pF





NOTES:
1. Perform test in accordance with test table; each output is tested separately.
2. All input and output cables are equal lengths of 50 ohm coaxial cables. Wire length should be ≤ .250 (6.35 mm) from t_p in to input pin and t_p out to output pin.
3. Outputs not under test connected to a 100 ohm resistor to ground.
4. Note that observed pulse amplitude is attenuated by one half.

FIGURE 6. Synchronous switching test circuit for device types 01 and 02.

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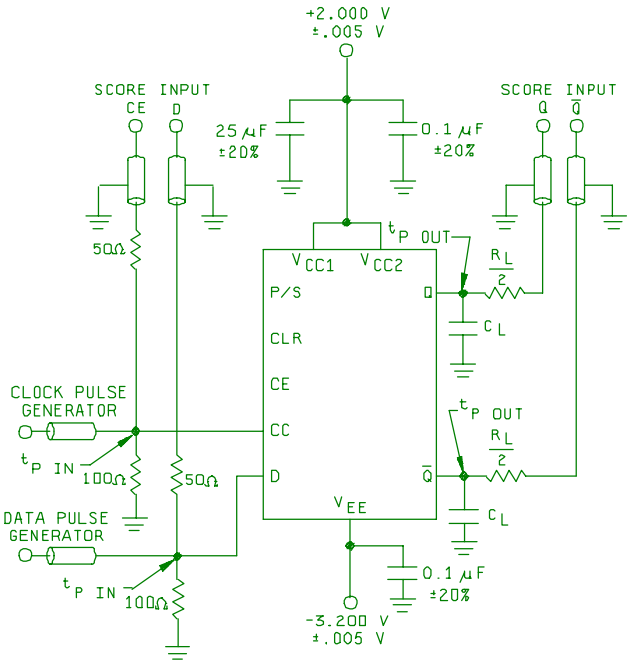
GENERATOR CHARACTERISTICS

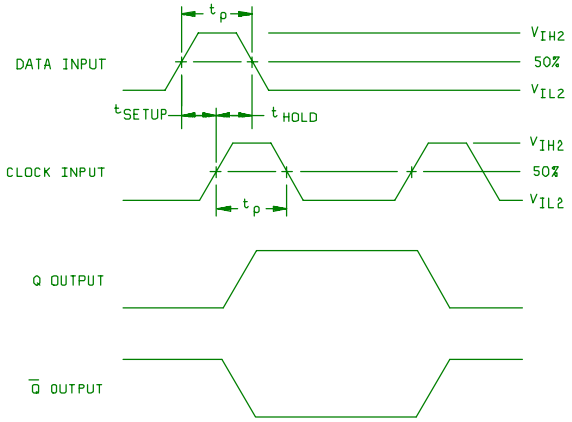
T_C	V_{IH2} ± 10 mV	V_{IL2} ± 10 mV
25 °C	+1.11 V	+0.31 V
125 °C	+1.24 V	+0.36 V
-55 °C	+1.01 V	+0.28 V

Z_{OUT} = 50Ω
t_p (data) = 40 ns
t_p (clock) = 40 ns
PRR = 1 MHz
Device type 01
T_{THL} = 2.0 ns (20%-80%)
T_{TLH} = 2.0 ns (20%-80%)
Device type 02
T_{THL} = 1.5 ns (20%-80%)
T_{TLH} = 1.5 ns (20%-80%)

$\frac{R_L}{2}$ = 50Ω ± 5%

SCOPE INPUT = 50Ω to GND
C_L (test jig) ≤ 5 pF





NOTES:
1. Perform test in accordance with test table; each output is tested separately.
2. All input and output cables are equal lengths of 50 ohm coaxial cables. Wire length should be ≤ .250 (6.35 mm) from t_p in to input pin and t_p out to output pin.
3. Outputs not under test connected to a 100 ohm resistor to ground.
4. Note that observed pulse amplitude is attenuated by one half.

FIGURE 7. Setup and hold test circuit for device types 01 and 02.

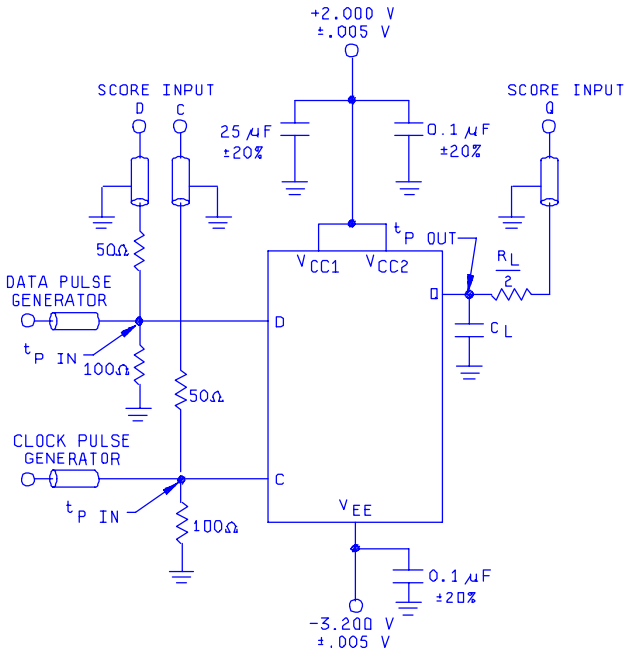
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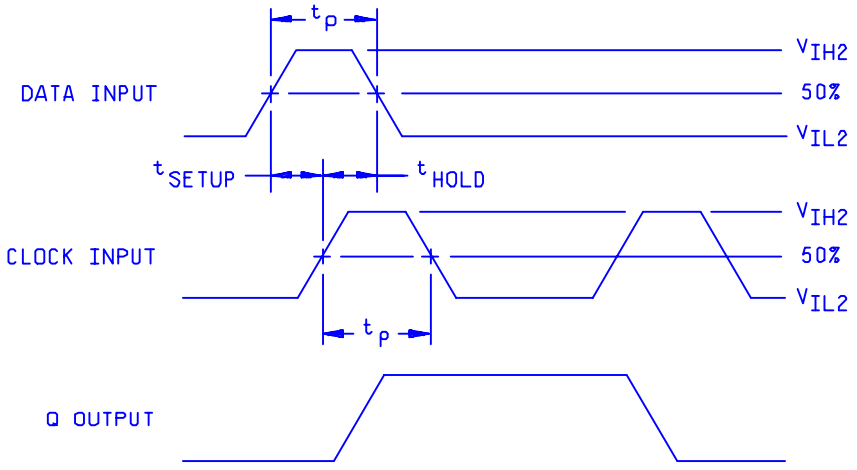
GENERATOR CHARACTERISTICS

T_C	V_{IH2} ±10 mV	V_{IL2} ±10 mV
25°C	+1.11 V	+0.31 V
125°C	+1.24 V	+0.36 V
-55°C	+1.01 V	+0.28 V

Z_{OUT} = 50Ω
t_{THL} = 2.0 ns (20% - 80%)
t_{TLH} = 2.0 ns (20% - 80%)
t_P (data) = 40 ns
t_P (clock) = 40 ns
PRR= 1 MHz

$R_L/2$ = 50Ω ± 5%
SCOPE INPUT = 50Ω to GND
C_L (test jig) ≤ 5 pF





NOTES
1. Perform test in accordance with test table; each output is tested separately.
2. All input and output cables are equal lengths of 50Ω coax cables. Wire length should be ≤ .250 (6.35 mm) from t_p in to input pin and t_p out to output pin.
3. Outputs not under test connected to a 100 ohm resistor to ground.
4. Note that observed pulse amplitude is attenuated by one half.

FIGURE 8. Setup and hold test circuit for device type 03.

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